

Agentic AI Engineering

Fundamentals — Quiz Set (30 Questions)

Chapter-weighted and tiered: Recall · Application · Analysis. Full answer key explains every correct answer and why each distractor is wrong.

Tier	Count	What it tests
Recall	7	Definitions and facts
Application	10	Causal mechanism in a scenario
Analysis	13	Trade-offs, failure modes, comparisons

Answer the 30 questions first, then check the answer key that follows. Each question is tagged with its chapter and tier.

Questions

1. An agent calls an API, it fails, and the agent tries an alternative approach. Which stage detected the failure and enabled the change? [CH 5 · APPLICATION]
 - A. Perceive
 - B. Act
 - C. Observe
 - D. Think
2. Why do production agents define explicit stopping conditions? [CH 5 · ANALYSIS]
 - A. To make responses more creative
 - B. To prevent infinite loops and ensure bounded, predictable execution
 - C. To increase the context window
 - D. To avoid using tools
3. What is 'function calling' in an agentic system? [CH 6 · RECALL]
 - A. The LLM generating prose for a human to read
 - B. The LLM emitting a structured request that external software executes
 - C. A way to fine-tune the model on new data
 - D. A method for chunking documents
4. An agent keeps choosing the wrong tool between 'search_invoices' and 'search_contracts'. What is the most likely root cause? [CH 6 · APPLICATION]
 - A. The context window is too small
 - B. The tool descriptions are vague or overlapping
 - C. The temperature is set too low
 - D. The vector database is misconfigured
5. Which memory type stores long-term facts, preferences, and stable knowledge? [CH 7 · RECALL]
 - A. Sensory memory
 - B. Working memory
 - C. Episodic memory
 - D. Semantic memory
6. A stakeholder says 'the agent should remember our refund policy.' What does this almost always mean in practice? [CH 7 · ANALYSIS]
 - A. Fine-tune the LLM on the policy
 - B. Store the policy in external memory and retrieve it when relevant
 - C. Increase the temperature
 - D. Add more tools

7. What is the correct order of the RAG pipeline? [CH 8 · RECALL]

- A. Retrieval → Indexing → Chunking → Generation
- B. Data Preparation (chunking) → Indexing (embeddings) → Retrieval → Generation
- C. Generation → Retrieval → Chunking → Indexing
- D. Embedding → Fine-tuning → Retrieval → Generation

8. An enterprise's policies change weekly. Why is RAG preferred over fine-tuning to keep answers current? [CH 8 · APPLICATION]

- A. RAG makes the model larger
- B. RAG decouples knowledge from the model — you update a knowledge base, not the weights
- C. Fine-tuning is faster than retrieval
- D. RAG removes the need for embeddings

9. Why does RAG reduce hallucinations? [CH 8 · ANALYSIS]

- A. It lowers the temperature to zero
- B. It grounds the answer in retrieved evidence instead of relying on model memory
- C. It increases the model's parameter count
- D. It disables the context window

10. Why must indexing and querying use the same embedding model? [CH 9 · ANALYSIS]

- A. To save storage space
- B. Because different models produce incompatible coordinate systems, wrecking retrieval
- C. To reduce temperature
- D. To enable fine-tuning

11. What does HNSW (Approximate Nearest Neighbor) trade to achieve speed at scale? [CH 10 · ANALYSIS]

- A. A small amount of accuracy for much faster search
- B. Security for speed
- C. Memory for correctness
- D. Nothing — it is exact and instant

12. Which best describes the Model Context Protocol (MCP)? [CH 11 · APPLICATION]

- A. A replacement for all APIs
- B. An open standard for how agents discover and use tools/resources through a consistent interface
- C. A new LLM architecture
- D. A vector database engine

13. A SOC analyst agent must gather evidence from multiple systems mid-reasoning during a live investigation. Which reasoning pattern fits best? [CH 12 · APPLICATION]

- A. Chain of Thought
- B. ReAct
- C. Tree of Thoughts
- D. Plan-and-Execute

14. Which reasoning pattern has the model critique its own output and self-correct before finalizing? [CH 12 · RECALL]

- A. Chain of Thought
- B. Reflection
- C. ReAct
- D. Plan-and-Execute

15. In the most common multi-agent architecture, what is the orchestrator's role? [CH 13 · RECALL]

- A. It writes all the code itself
- B. It decomposes work, assigns specialists, monitors progress, and combines results
- C. It stores the vector embeddings
- D. It replaces the LLM

16. According to the 'earn the complexity' principle, when should you add more agents? [CH 13 · ANALYSIS]

- A. Immediately, for every project
- B. Only when specialization creates measurable value
- C. Whenever the LLM is large enough
- D. Never — one agent is always best

17. A team needs enterprise orchestration with loops, branching, persistent state, and human-approval checkpoints. Which framework fits best? [CH 14 · APPLICATION]

- A. LangChain
- B. LangGraph
- C. LlamaIndex
- D. AutoGen

18. What distinguishes Evaluator–Optimizer from Reflection? [CH 15 · ANALYSIS]

- A. Evaluator–Optimizer uses an independent reviewer; Reflection is self-review by the same agent
- B. They are identical
- C. Reflection uses two agents; Evaluator–Optimizer uses one
- D. Evaluator–Optimizer removes human oversight

19. A workflow must NOT execute a high-consequence, hard-to-reverse action without sign-off. Which pattern applies? [CH 15 · APPLICATION]

- A. Event-Driven
- B. Human-in-the-Loop (HITL)
- C. Router
- D. State Machine

20. A retrieved web page contains a hidden instruction telling the agent to leak data. What attack is this? [CH 16 · APPLICATION]

- A. Direct prompt injection
- B. Indirect prompt injection
- C. Denial of service
- D. SQL injection

21. What is the core idea of 'Defense in Depth' for enterprise AI? [CH 16 · ANALYSIS]

- A. One strong control (a firewall) is enough
- B. Protection lives at every layer, and each layer assumes another may eventually fail
- C. Only the model needs protection
- D. Guardrails replace identity controls

22. Which are the three pillars of AI observability? [CH 17 · RECALL]

- A. Logs, Metrics, Traces
- B. Prompts, Tools, Memory
- C. Accuracy, Latency, Cost
- D. Detect, Contain, Remediate

23. What is 'red teaming' in the context of production AI? [CH 17 · APPLICATION]

- A. Load testing the API
- B. Deliberately attacking your own system (prompt injection, policy violations) to find weaknesses first
- C. A/B testing prompts
- D. Monitoring uptime

24. Why is the system/developer/user prompt hierarchy also a security boundary? [CH 18 · ANALYSIS]

- A. It makes prompts shorter
- B. It prevents a crafted user prompt from overriding organizational policy
- C. It increases model temperature
- D. It removes the need for guardrails

25. What problem does enterprise AI architecture primarily prevent? [CH 19 · ANALYSIS]

- A. Model hallucination
- B. AI silos — fragmented, duplicated, inconsistent per-team deployments
- C. Slow inference
- D. Token overflow

26. Per the chapter, what is the most common reason great AI products fail? [CH 20 · ANALYSIS]

- A. The model is too small
- B. They start from a model ('what can we build?') instead of a real problem
- C. They use too many guardrails
- D. They deploy on the cloud

27. For AI-assisted vulnerability management, what improves prioritization beyond raw CVSS scores? [CH 21 · APPLICATION]

- A. Alphabetical ordering
- B. Business context — exposure, asset criticality, and reachability
- C. Random sampling
- D. Only the CVE publication date

28. What is the primary value of agentic AI in quantitative finance? [CH 22 · ANALYSIS]

- A. Perfectly predicting future prices
- B. Improving decision quality under uncertainty by processing more information consistently
- C. Eliminating all market risk
- D. Replacing human portfolio managers entirely

29. A model's performance degrades over months with NO code changes. What is the most likely cause? [CH 23 · ANALYSIS]

- A. A syntax error
- B. Drift — the environment (data, relationships, or behavior) changed
- C. The context window shrank
- D. A missing import

30. What does the handbook describe as the ultimate objective of agentic AI? [CH 24 · RECALL]

- A. The largest possible model
- B. Intelligent organizations that learn and adapt, not merely intelligent software
- C. Fully autonomous systems with no humans
- D. Eliminating all knowledge bases

Answer Key & Explanations

Each answer explains the causal *why* behind the correct option and why each distractor is wrong.

1. Correct: C [CH 5 · APPLICATION]

Why: Observe evaluates the result of an action (success, error, new information) and closes the feedback loop, which is what lets the agent detect failure and adapt.

Distractors: A — Perceive gathers initial inputs; it does not evaluate an action's outcome. · B — Act performs the API call — it does not judge whether it worked. · D — Think chooses the next action, but detection of the failure happens in Observe first.

2. Correct: B [CH 5 · ANALYSIS]

Why: Without stopping conditions an agent could loop forever; every production system bounds execution (max steps, budget, success criteria) for predictable behavior.

Distractors: A — Creativity is controlled by temperature, not stopping conditions. · C — Stopping conditions do not change context window size. · D — Stopping conditions bound the loop; they do not prevent tool use.

3. Correct: B [CH 6 · RECALL]

Why: Function calling lets the model emit a structured request (name + arguments) that specialized software runs, separating reasoning (model decides) from execution (software does).

Distractors: A — That describes ordinary text generation, not function calling. · C — Fine-tuning changes the model's weights — unrelated to function calling. · D — Chunking is a RAG data-prep step, not function calling.

4. Correct: B [CH 6 · APPLICATION]

Why: The model selects tools based entirely on their descriptions; vague or overlapping descriptions are the leading cause of wrong-tool errors — tool descriptions are effectively prompt engineering.

Distractors: A — A small context window causes truncation, not systematic wrong-tool selection between two clear tools. · C — Low temperature makes output more deterministic, which would not cause wrong-tool selection by itself. · D — A vector DB issue affects retrieval quality, not tool selection.

5. Correct: D [CH 7 · RECALL]

Why: Semantic memory holds stable long-term knowledge ('What do I know?'), as opposed to events (episodic) or transient processing (sensory/working).

Distractors: A — Sensory memory is the shortest — just-perceived, unprocessed input. · B — Working memory is what's actively in use for the current problem. · C — Episodic memory stores events the agent experienced.

6. Correct: B [CH 7 · ANALYSIS]

Why: Production systems separate reasoning from storage: 'remember X' means storing X externally and retrieving it — not retraining the model, which is expensive and brittle.

Distractors: A — Fine-tuning to store a fact is the classic expensive, brittle anti-pattern the chapter warns against. · C — Temperature controls output variability, not memory. · D — Tools enable actions, not persistent factual recall.

7. Correct: B [CH 8 · RECALL]

Why: RAG runs Data Preparation/chunking → Indexing (embeddings into a vector DB) → Retrieval (semantic search) → Generation (LLM answers from retrieved chunks).

Distractors: A — Retrieval cannot precede indexing — there would be nothing to retrieve. · C — Generation is last, not first. · D — Fine-tuning is not part of the RAG pipeline; RAG adds knowledge without changing the model.

8. Correct: B [CH 8 · APPLICATION]

Why: RAG supplies knowledge at inference time, so updating the knowledge base instantly updates answers — no costly, frequent retraining.

Distractors: **A** — RAG does not change model size. · **C** — Retraining weekly is impractical and expensive; retrieval is the cheaper path. · **D** — RAG depends on embeddings for semantic retrieval.

9. Correct: B [CH 8 · ANALYSIS]

Why: Hallucinations occur when the model must predict text without sufficient information; supplying grounded, retrieved context lets it reason from real evidence.

Distractors: **A** — Temperature is a separate control and not how RAG reduces hallucination. · **C** — Parameter count is unrelated to retrieval grounding. · **D** — RAG uses the context window to inject retrieved chunks — it does not disable it.

10. Correct: B [CH 9 · ANALYSIS]

Why: If documents and queries are embedded by different models, their vector spaces are incompatible and similarity comparisons become meaningless — retrieval quality collapses.

Distractors: **A** — Storage is not the reason; correctness is. · **C** — Temperature is unrelated to embeddings. · **D** — Consistency is about retrieval quality, not fine-tuning.

11. Correct: A [CH 10 · ANALYSIS]

Why: ANN algorithms like HNSW navigate a similarity graph into the relevant neighborhood instead of comparing every vector, sacrificing a tiny bit of accuracy for a large speed gain.

Distractors: **B** — ANN does not trade away security. · **C** — It is about accuracy vs. speed, not memory vs. correctness. · **D** — Exact search compares everything and is slow; ANN is approximate by design.

12. Correct: B [CH 11 · APPLICATION]

Why: MCP standardizes how agents discover and use external tools and resources (Host → Client → Server → API); it does not replace APIs — it standardizes access to them.

Distractors: **A** — MCP does not replace APIs; APIs remain the mechanism software uses to communicate. · **C** — MCP is a protocol, not a model architecture. · **D** — MCP is not a vector database.

13. Correct: B [CH 12 · APPLICATION]

Why: ReAct alternates reasoning with external action (Think → Act → Observe), letting the agent gather new information mid-reasoning — ideal for changing, real-time environments.

Distractors: **A** — Chain of Thought is purely internal — it cannot search or retrieve mid-reasoning. · **C** — Tree of Thoughts explores branches but is oriented to comparing options, not live evidence-gathering. · **D** — Plan-and-Execute suits stable objectives, not rapidly changing investigations.

14. Correct: B [CH 12 · RECALL]

Why: Reflection has the model review and critique its own reasoning, then self-correct identified mistakes before producing the final answer.

Distractors: **A** — Chain of Thought reasons in steps but does not self-critique a finished answer. · **C** — ReAct alternates reasoning and action. · **D** — Plan-and-Execute strategizes then executes; it is not self-critique.

15. Correct: B [CH 13 · RECALL]

Why: The Orchestrator–Worker pattern uses a project-manager-like orchestrator that breaks work into tasks, assigns specialists, monitors, and synthesizes the final result.

Distractions: **A** — Specialists do the work; the orchestrator coordinates. · **C** — Embedding storage is the vector DB's job. · **D** — The orchestrator is itself an agent powered by an LLM; it doesn't replace it.

16. Correct: B [CH 13 · ANALYSIS]

Why: Start with one capable agent; add specialists only when specialization delivers measurable value, because more agents add coordination overhead, cost, and latency.

Distractions: **A** — Defaulting to multi-agent is the most common over-engineering mistake. · **C** — Model size does not justify adding agents. · **D** — Multi-agent systems are valuable when justified — 'never' is too absolute.

17. Correct: B [CH 14 · APPLICATION]

Why: LangGraph represents workflows as graphs (loops, branching, retries, persistent state, HITL checkpoints), making it a preferred enterprise orchestration framework.

Distractions: **A** — LangChain's linear chains get hard to manage for complex, stateful workflows. · **C** — LlamaIndex specializes in retrieval/RAG, not orchestration. · **D** — AutoGen focuses on conversational collaboration/research, not production orchestration with state.

18. Correct: A [CH 15 · ANALYSIS]

Why: Reflection is self-review (can share the generator's blind spots); Evaluator–Optimizer externalizes review to an independent agent with its own criteria, catching errors the author cannot see.

Distractions: **B** — They achieve the same goal by different means, not identical mechanisms. · **C** — This reverses the two — Reflection is self (one agent); Evaluator–Optimizer adds a second. · **D** — Neither is about removing human oversight.

19. Correct: B [CH 15 · APPLICATION]

Why: HITL gives the AI a recommendation role for high-consequence decisions (approving loans, containing incidents, deploying software) while a human retains authority and accountability.

Distractions: **A** — Event-Driven is about what triggers a workflow, not approval gating. · **C** — Router classifies and directs requests; it does not gate high-risk actions. · **D** — State machines add auditability but don't inherently require human sign-off.

20. Correct: B [CH 16 · APPLICATION]

Why: Indirect prompt injection hides malicious instructions in external sources (web pages, documents, knowledge bases) that the user never sees but the AI ingests — turning trusted retrieval into an attack vector.

Distractions: **A** — Direct injection is in the user's own input, not a retrieved document. · **C** — DoS targets availability, not reasoning. · **D** — SQL injection targets a database query, not the model's reasoning.

21. Correct: B [CH 16 · ANALYSIS]

Why: Defense in Depth layers protection across identity, knowledge, memory, reasoning, tools, infrastructure, monitoring, and governance — no single safeguard is expected to stop every attack.

Distractions: **A** — A single control is exactly what Defense in Depth rejects. · **C** — Reasoning, memory, retrieval, tools, and infrastructure all need protection. · **D** — Guardrails and least-privilege identity are complementary, not substitutes.

22. Correct: A [CH 17 · RECALL]

Why: Logs (what happened), Metrics (numeric trends), and Traces (a request's full lifecycle) are the three pillars of observability.

Distractors: B — Those are agent components, not observability pillars. · C — Those are individual metrics, not the three pillars. · D — That is the incident-response sequence (Ch 23).

23. Correct: B [CH 17 · APPLICATION]

Why: Red teaming is the AI equivalent of penetration testing — evaluators deliberately challenge the system with adversarial inputs to surface weaknesses before real adversaries do.

Distractors: A — Load testing measures capacity, not adversarial robustness. · C — A/B testing compares variants; it is not adversarial. · D — Uptime monitoring is availability, not adversarial testing.

24. Correct: B [CH 18 · ANALYSIS]

Why: Each layer narrows scope; maintaining the hierarchy stops a user request from overriding permanent organizational policy — the structural defense against prompt injection.

Distractors: A — The hierarchy is about precedence, not length. · C — It is unrelated to temperature. · D — It complements guardrails; it does not replace them.

25. Correct: B [CH 19 · ANALYSIS]

Why: Deploying many independent agents with different prompts, models, and controls creates AI silos that raise cost, risk, and inconsistency; standardization and reusable shared services prevent this.

Distractors: A — Hallucination is addressed by RAG/grounding, not architecture per se. · C — Inference speed is an infrastructure/cost concern, not the primary architectural problem. · D — Token overflow relates to context management, not silos.

26. Correct: B [CH 20 · ANALYSIS]

Why: Technology should support business objectives, not define them; starting with the model produces impressive demonstrations searching for customers — technical success with zero adoption.

Distractors: A — Model size is rarely the cause; solving the wrong problem is. · C — Guardrails support safe deployment; they don't cause product failure. · D — Cloud deployment is normal and not a failure cause.

27. Correct: B [CH 21 · APPLICATION]

Why: A critical vuln on an isolated lab system may pose less real risk than a medium vuln on a public-facing financial app; reasoning over business context improves prioritization.

Distractors: A — Alphabetical ordering ignores risk entirely. · C — Random sampling is not prioritization. · D — Publication date alone ignores exploitability and exposure.

28. Correct: B [CH 22 · ANALYSIS]

Why: Markets are information-processing systems with irreducible uncertainty; AI supports better, more consistent decisions (probability, risk, expected value) rather than promising prediction certainty.

Distractors: A — The chapter explicitly rejects prediction as the goal. · C — Risk cannot be eliminated; it is managed. · D — Human oversight remains essential in regulated finance.

29. Correct: B [CH 23 · ANALYSIS]

Why: AI systems change even when developers do nothing: data drift, concept drift, or behavioral drift degrade performance as the environment evolves — which is why continuous drift detection is essential.

Distractors: **A** — A syntax error would break the system immediately, not degrade it slowly. · **C** — Context window size does not change on its own. · **D** — A missing import causes an immediate failure, not gradual decay.

30. Correct: B [CH 24 · RECALL]

Why: The goal reframes from intelligent software to intelligent organizations — ones that learn continuously, share knowledge, and improve, with AI amplifying reasoning as infrastructure.

Distractors: **A** — Model size is a means, never the objective. · **C** — The future is explicitly not full human replacement. · **D** — Knowledge bases are foundational, not eliminated.